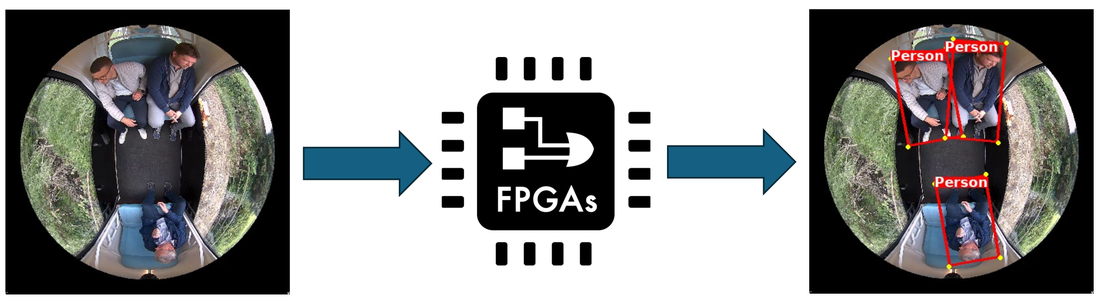
FINN4YOLO: Unlocking Real-Time Fisheye Vision with FINN-Accelerated YOLO on FPGA

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Motivation

- Fisheye cameras provide an ultra-wide field of view, making them attractive for autonomous robots, surveillance, and intelligent transportation systems.
- Fisheye image processing can benefit from edge deployment to reduce latency and bandwidth requirements while improving data privacy.
- Key challenges for real-time edge deployment include:
  - Fisheye cameras introduce geometric distortion in the images
  - Oriented object detection models have high computational cost
- **Goal: Enable efficient real-time oriented object detection on FPGAs for fisheye vision, particularly in moving platforms such as trains.**



Contribution

- We develop FINN-compatible oriented bounding-box (OBB) detection architectures for FPGA deployment.
- We adapt YOLOv3n-tiny, YOLOv5n, and YOLOv8n to address FINN compilation and hardware constraints.
- We evaluate detection accuracy across four fisheye datasets and benchmark FPGA performance on ZCU104 and Alveo U250 platforms.

Methodology and Key Ideas

- Development of customized quantized OBB detection heads for FINN-compatible FPGA deployment:
  - YOLOv3n-tiny-OBB: Extended output channels to predict an orientation angle for each anchor while retaining the original backbone and neck.
  - YOLOv5n-OBB: Integrated the OBB head and removed skip connections to simplify the computation graph for FINN compilation.
  - YOLOv8n-OBB: Replaced SiLU activations with ReLU to ensure FINN compatibility.
- Quantization-aware training (QAT) using standard YOLO augmentations at an input resolution of 416 × 416 × 3.
- The extended FINN-plus compiler is used, integrating custom operations into the streamlining and hardware-layer conversion steps to enable YOLO model compilation.
- Hardware-aware FPGA compilation with automatic FIFO sizing and a target clock period of 10 ns.
- On-board evaluation of the deployed models for accuracy, throughput, resource utilization, and power consumption.

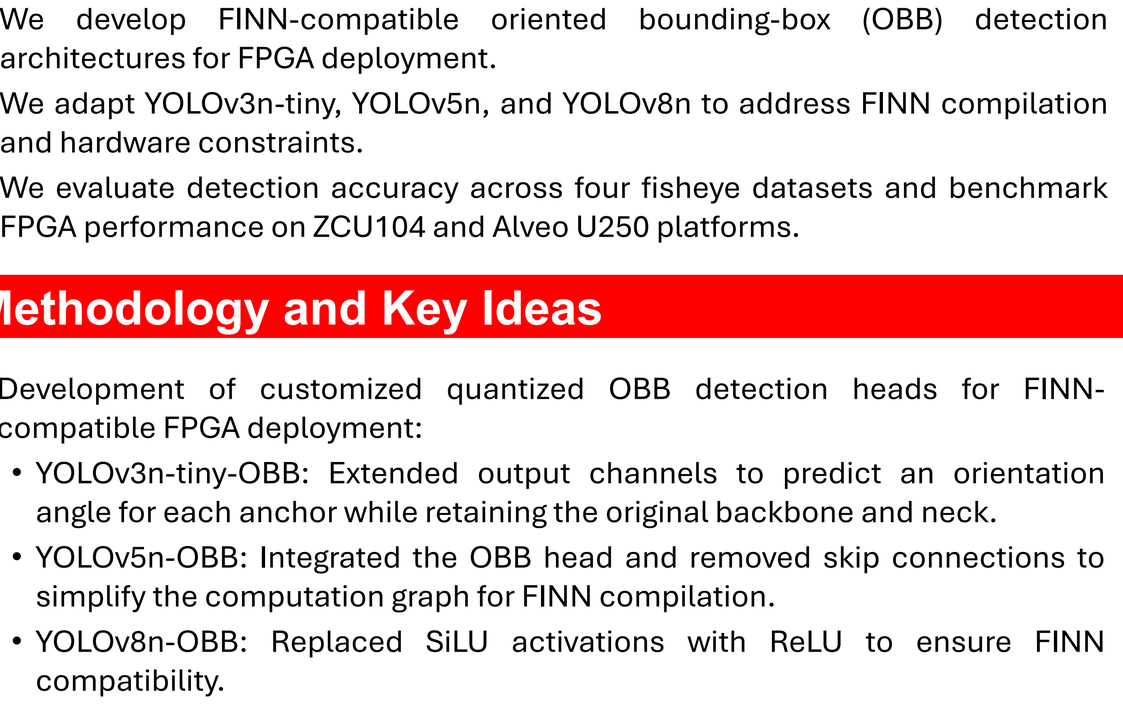


Figure 1: FINN-based workflow for training, compilation, FPGA deployment, and on-board evaluation of OBB YOLO models

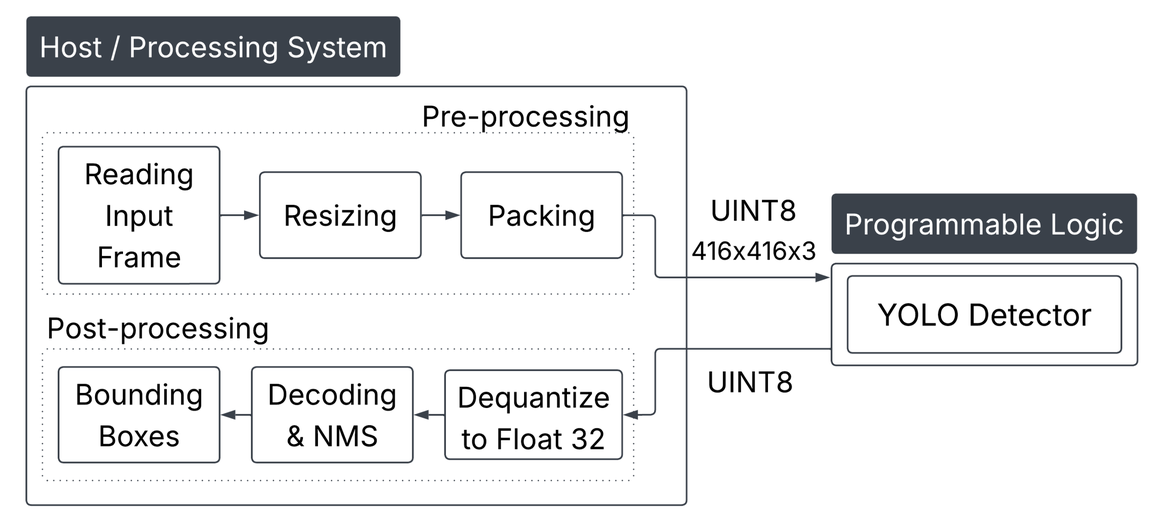


Figure 2: End-to-end FPGA inference pipeline comprising host-side preprocessing and post-processing and PL-based YOLO inference.

Results

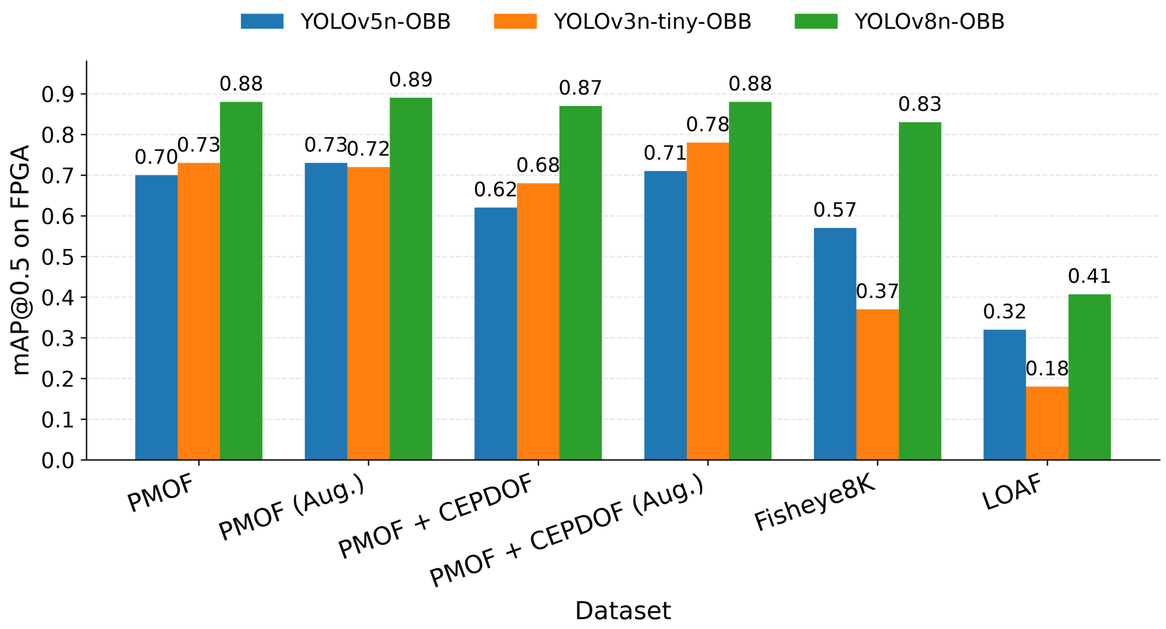


Figure 3: Detection accuracy of the models on FPGA across fisheye datasets.

Model	Board	LUT	FF	BRAM	DSP	Power (W)
YOLOv3n-tiny-OBB	ZCU104	22.95 %	10.12 %	60.80 %	1.79 %	3.90
YOLOv3n-tiny-OBB	U250	13.28 %	7.57 %	11.74 %	0.18 %	17.98
YOLOv5n-OBB	U250	23.50 %	14.15 %	29.78 %	20.05 %	22.30
YOLOv8n-OBB	U250	28.59 %	16.75 %	43.63 %	17.94 %	23.84

Table 1: Resource utilization and on-chip power consumption across models and FPGA platforms.

The evaluated architectures demonstrate distinct accuracy–throughput–resource trade-offs, reaching up to 234 FPS on the Alveo U250.

Discussion

- YOLOv3n-tiny-OBB delivers the highest throughput: 234 FPS on U250 and 42 FPS on ZCU104, with 0.74 mAP@0.5 on PMOF.
- YOLOv5n-OBB provides a balanced design point (62 FPS, 0.71 mAP@0.5), while YOLOv8n-OBB favors accuracy at higher resource cost and lower throughput (30 FPS).
- Overall, the models expose clear accuracy–throughput–resource–power trade-offs, enabling application-specific FPGA deployment.

Future Work

- Deployment of YOLOv11n-OBB and YOLO26n-OBB on FPGA using the FINN toolchain.
- Evaluate the performance of the deployed models across all six datasets in terms of latency, FPS, and on-chip power consumption.
- Explore resolution and quantization trade-offs for improved efficiency.

References

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